

Reallocation of the Women's Workforce within the Organised Manufacturing Sector: Evidence from the ASI data

Laxmikanta Sarangi*¹ and Sulata Pradhan²

¹Statistical Officer, Directorate of Economics and Statistics (DE&S), Govt. Of Odisha

²Department of Economics, South Asian University, New Delhi, M.A (Economics)

Abstract

India's industrial policy increasingly treats manufacturing as an engine of employment, yet women remain a small and unevenly distributed part of the organised manufacturing workforce. This paper asks whether the changing presence of women in organised manufacturing has been driven by what happens *inside* industries or by the movement of employment *across* them. Using industry-level data from the Annual Survey of Industries for 2014–2024, we decompose the growth of the female share of manufacturing employment into a within-industry effect and a between-industry (reallocation) effect, following the structural-change framework of Ocampo et al. (2009) as applied by Tejani and Kucera (2021). The within-industry component dominates throughout the period and is markedly volatile, including a pronounced spike in 2021, while the reallocation component stays small. Changes in women's employment shares thus originate mainly in processes internal to industries rather than in a structural shift of employment toward female-intensive activities. A complementary cross-section shows a weak but negative association between capital intensity and the female employment share: more capital-intensive industries employ proportionally fewer women. Read together, the results indicate that capital deepening and technological upgrading have not, on their own, widened women's access to manufacturing jobs. We caution that share-based gains can also reflect faster male job loss, and we draw out the implications for skilling and anti-segregation policy.

Keywords: female labour reallocation; defeminization; organised manufacturing; within-industry decomposition; capital intensity; India

JEL classification: J16; J21; L60; O14

*Corresponding author. Email: [corresponding-author email -- lks19.11.12@gmail.com.

Contents

1	Introduction	3
2	Literature Review	4
3	Data and Variables	6
4	Methodology	6
5	Results	8
5.1	Within-industry versus reallocation effects	8
5.2	Capital intensity and the female employment share	9
6	Discussion	11
7	Conclusion	12
8	Limitations and Future Work	12

1 Introduction

In the analysis of economic development, structural transformation is understood as the movement of labour from low-productivity to high-productivity activities. Manufacturing occupies a central place in this process because of its capacity to generate employment, foster technological progress, and sustain economic growth. The benefits of industrial growth have not, however, been distributed evenly across genders, and the extent to which women have been drawn into structural transformation remains a subject of considerable debate.

Many developing countries saw a surge of female employment in labour-intensive industries during their export-oriented industrialisation. Later evidence complicated that picture: the relationship between industrialisation and women’s employment turned out to be neither universal nor linear. Technological upgrading, rising capital intensity, and changing production structures have often gone together with a decline in women’s share of manufacturing employment, giving rise to the phenomenon of “defeminization.” Such episodes suggest that the dynamics of women’s employment in manufacturing are shaped not only by overall economic growth but also by changes occurring within industries and by the movement of labour across industries.

In India, schemes such as Make in India and the Production Linked Incentive programme envisage manufacturing as a major source of employment and a driver of economic expansion. Raising women’s participation in economic activity has, at the same time, become an explicit policy objective. Despite these efforts, women’s employment in manufacturing remains limited and heavily concentrated in traditional industries such as textiles, apparel, and food processing. Their representation in technologically advanced and high-growth industries is comparatively low, pointing to persistent occupational segregation and unequal access to emerging employment opportunities.

A substantial literature has examined female labour force participation, gender wage differentials, and the feminization and defeminization of employment. Far less attention has gone to the reallocation of female labour *within* the manufacturing sector. In particular, there is limited evidence on whether changes in female employment are driven mainly by shifts within industries or by the movement of labour across industries. Existing studies are largely cross-country in design and say little about the specific experience of India’s organised manufacturing sector.

Given the weight manufacturing carries in India’s growth strategy and the stated goal of inclusive structural transformation, the dynamics of female labour reallocation deserve closer attention. Tracing how the distribution of women workers across manufacturing industries has changed can reveal which activities have widened, and which have narrowed, women’s employment opportunities. Such an examination is necessary to judge whether industrial transformation has fostered greater gender inclusion or has instead reinforced existing patterns of segregation.

This study examines the pattern of female labour reallocation in India’s organised manufacturing sector using industry-level data from the Annual Survey of Industries. By decomposing changes in female employment into within-industry and between-industry effects, it identifies the relative contribution of structural change and industry-specific dynamics to the evolution

of women’s employment.

2 Literature Review

Early studies argued that export-oriented industrialisation set off a feminization of labour: female employment in manufacturing rose because women were a comparatively cheap and available source of labour. Elson and Pearson (1981) showed that multinational firms in labour-intensive industries such as garments and electronics preferred female workers, whom they regarded as more disciplined and less likely to unionise. Building on this, Standing (1989, 1999) advanced the notion of a “global feminization of labour,” arguing that globalisation and intensifying competition encouraged the substitution of male labour by cheaper female labour. The spread of globally integrated production networks both drew large numbers of women into export manufacturing and shaped the terms on which they were employed (Joekes, 1999; Barrientos et al., 2004).

The orthodox trade account explains this through the Stolper–Samuelson theorem: trade liberalisation raises demand for a country’s abundant factor, which in developing economies is taken to include relatively low-skilled female labour. Heterodox approaches instead stress that firms exploit gender wage differentials to preserve competitiveness (Seguino, 1997, 2000). Although these studies established a positive association between globalisation and women’s employment, later evidence suggested that the relationship was neither universal nor permanent.

The contribution of Kucera and Tejani (2014) shifted the debate from simple increases in female employment to changes in the female *share* of manufacturing employment, formalising the concepts of feminization and defeminization. Using decomposition analysis for 36 countries between 1981 and 2008, they found that both feminization and defeminization were driven predominantly by within-industry change rather than by inter-industry reallocation. Contrary to the proposition associated with Caraway (2007)—that structural shifts toward capital-intensive industries explain declining female employment—they found that labour-intensive industries accounted for the largest contribution to changes in female employment shares in 25 of 36 countries, and that within-industry effects dominated in 33. Textiles and apparel emerged as the principal drivers of both feminization and defeminization. In East Asian economies such as South Korea and Taiwan, falling female shares within these industries coincided with technological upgrading, a sign that structural transformation need not automatically benefit women. The literature thus came to recognise that changes in women’s employment are determined not only by sectoral shifts but also by transformations occurring inside industries.

Tejani and Milberg (2016) revisited the feminization hypothesis by asking whether export expansion or technological change better accounts for changes in female manufacturing employment. Estimating a fixed-effects model for developing countries, they found that capital intensity and labour-productivity growth were negatively associated with the female share of employment, whereas export growth itself had no statistically significant effect. Their results challenged the stylised fact of a universal feminization and pointed instead to an anti-female bias in labour demand as industries upgrade: as production becomes more capital-intensive and

technologically advanced, employers tend to favour male workers. Globalisation alone, on this reading, does not determine women’s employment outcomes; the technological conditions of production are decisive—a conclusion of particular relevance for developing countries pursuing industrial upgrading. In a related vein, Greenstein and Anderson (2017) connect defeminization to premature deindustrialisation, arguing that when manufacturing employment peaks at lower income levels the female-intensive phase of industrialisation is correspondingly foreshortened.

A further strand of the literature concerns occupational segregation. Women remain concentrated in labour-intensive, low-value-added industries—textiles, garments, and food processing in particular. Kucera and Tejani (2014) caution that although textiles and apparel are traditionally female-intensive, the link between labour intensity and female intensity is not straightforward and the correlation between the two is weak. Gender segregation is reinforced by social norms and by employer perceptions of which occupations are suitable for women. The male bias that accompanies technological upgrading has been attributed to prevailing stereotypes, to differences in perceived skill, and to the persistence of the breadwinner ideology. Women are therefore over-represented in industries marked by low wages, limited skill upgrading, and precarious employment.

Beyond labour-market mechanics, feminist economists have criticised the unitary household model and proposed a bargaining framework in which household members hold unequal power determined by access to resources, employment opportunities, and social institutions. A woman’s bargaining power depends on her ownership of productive assets, her access to employment and communal resources, the support available from the state, and prevailing social norms. These conditions influence whether women participate in labour markets and which occupations they can enter. Labour-allocation patterns, on this view, are embedded in broader social and institutional structures rather than being purely market-driven, and women frequently bargain from a weaker position owing to limited assets and restrictive norms.

In India, women’s participation in manufacturing has stayed low even as female labour force participation has risen. Recent evidence by Mitra (2026) shows that women are concentrated in traditional industries such as textiles, apparel, and food processing—industries characterised by low wages and few opportunities for skill upgrading. Data from the Annual Survey of Industries and the Annual Survey of Unincorporated Sector Enterprises indicate that wearing apparel alone accounts for close to one-third of female employment in formal manufacturing and almost half of female employment in informal manufacturing. Occupational segregation has kept women out of emerging industries such as pharmaceuticals, electrical equipment, and rubber products, limiting their part in India’s industrial transformation. Despite Make in India and the Production Linked Incentive schemes, then, women’s integration into dynamic manufacturing activities remains constrained.

The literature on structural transformation generally decomposes employment change into two parts: within-industry effects, which reflect changes in employment shares inside industries, and between-industry (reallocation) effects, which reflect the movement of labour across industries. Kucera and Tejani (2014) showed that within-industry effects dominate changes in female employment across countries. Most existing work, however, studies changes in female shares

rather than the reallocation of female labour across industries. Studies of India, similarly, concentrate on female labour force participation, gender wage gaps, or occupational segregation, with little attention to how female workers are redistributed among manufacturing industries over time.

3 Data and Variables

There is a large literature on feminization, defeminization, bargaining power, and labour reallocation, but the reallocation of female labour *within* the manufacturing sector has drawn little attention. The decomposition exercises that do exist are generally conducted at a macro, cross-country level, and only a few studies examine female labour reallocation in India. A clear gap therefore remains in understanding how the distribution of female employment across industries within a sector responds to structural transformation in India’s organised manufacturing.

This study addresses that gap using data from the Annual Survey of Industries (ASI), the principal source of statistics on India’s registered (organised) manufacturing sector. The ASI covers factories registered under the Factories Act, 1948, and so describes the formal segment of manufacturing; the unorganised sector lies outside its frame. We use industry-level employment data—total employment and female employment, by industry, as classified in the ASI—over the ten years from 2014 to 2024. Because the decomposition operates on annual growth rates (Section 4), each year’s value is computed as the change from that year to the next; the resulting series therefore spans the base years 2014–2023 while drawing on ASI rounds through 2024. The female employment share is the ratio of female to total employment, defined for the manufacturing sector as a whole and at the level of individual industries.

4 Methodology

To examine the factors behind the reallocation of women’s manufacturing employment, the study uses the decomposition framework developed by Ocampo et al. (2009) and applied by Tejani and Kucera (2021). The method splits the growth of the female employment share into two components—a within-industry effect and a reallocation effect—thereby separating change that occurs inside industries from change that arises as employment shifts across industries.

The female share of manufacturing employment is the ratio of female employment to total manufacturing employment, F/L , where F denotes total female employment in manufacturing and L denotes total manufacturing employment (women and men combined). At the industry level the female employment share is F_i/L_i . Since aggregate manufacturing employment is the sum of employment across industries, the overall female share can be written as

$$\frac{F}{L} = \frac{\sum_i F_i}{\sum_i L_i}. \quad (1)$$

The growth of the female share in the manufacturing sector is decomposed as

$$g = \frac{1}{1 + \hat{L}} \left[\sum_i \theta_{0i} (\hat{F}_i - \hat{L}_i) + \sum_i (\theta_{0i} - \varepsilon_{0i}) \hat{L}_i \right], \quad (2)$$

where the terms are defined as

$$\hat{L}_i = \frac{L_{1i} - L_{0i}}{L_{0i}} \quad (\text{growth rate of total employment in industry } i), \quad (3)$$

$$\hat{F}_i = \frac{F_{1i} - F_{0i}}{F_{0i}} \quad (\text{growth rate of female employment in industry } i), \quad (4)$$

$$\theta_{0i} = \frac{F_{0i}}{F_0} \quad (\text{industry } i\text{'s share of total female employment in base year 0}), \quad (5)$$

$$\varepsilon_{0i} = \frac{L_{0i}}{L_0} \quad (\text{industry } i\text{'s share of total manufacturing employment in base year 0}), \quad (6)$$

$$\hat{L} = \frac{L_1 - L_0}{L_0} \quad (\text{aggregate growth rate of total manufacturing employment}). \quad (7)$$

We decompose the growth rates of female employment shares rather than the difference in those shares, and do so on annual data wherever possible rather than over period endpoints.

Within-industry effect

The within-industry component is

$$G_W = \sum_i \theta_{0i} (\hat{F}_i - \hat{L}_i). \quad (8)$$

For any industry, the within-industry effect is the difference between industry-level female employment growth and industry-level total employment growth, weighted by the industry's share of female employment relative to female employment in manufacturing as a whole. The effect is positive (negative) when industry-level female employment grows faster (slower) than industry-level total employment. Positive within-industry effects can also arise when men lose jobs at a higher rate than women—during a recession, for example. Such an outcome represents no real improvement for either women or men, so within-industry effects must be read in the broader context of the economy.

Reallocation effect

The reallocation component is

$$G_r = \sum_i (\theta_{0i} - \varepsilon_{0i}) \hat{L}_i. \quad (9)$$

For any industry, the reallocation effect is the difference between the industry's share of female employment (relative to female employment for manufacturing as a whole) and its share of total employment (relative to total employment for manufacturing as a whole), multiplied by industry-level employment growth. The term $(\theta_{0i} - \varepsilon_{0i})$ provides a measure of gender segregation, with smaller absolute values indicating less segregation. Positive reallocation effects arise

when employment grows in industries for which this difference is positive—that is, in industries with an above-average female share. They also arise, however, when employment *contracts* in industries with a below-average female share. As with the within-industry component, a positive reallocation effect need not signal genuine improvement for workers of either sex and must be interpreted in the broader country context. Taken together, the within-industry and reallocation components yield $g = (1 + \hat{L})^{-1} (G_W + G_r)$.

5 Results

5.1 Within-industry versus reallocation effects

Figure 1 presents the decomposition of changes in the female employment share in organised manufacturing into within-industry and reallocation effects over 2014–2023. Fluctuations in women’s employment shares were driven predominantly by within-industry dynamics rather than by shifts of employment across industries.

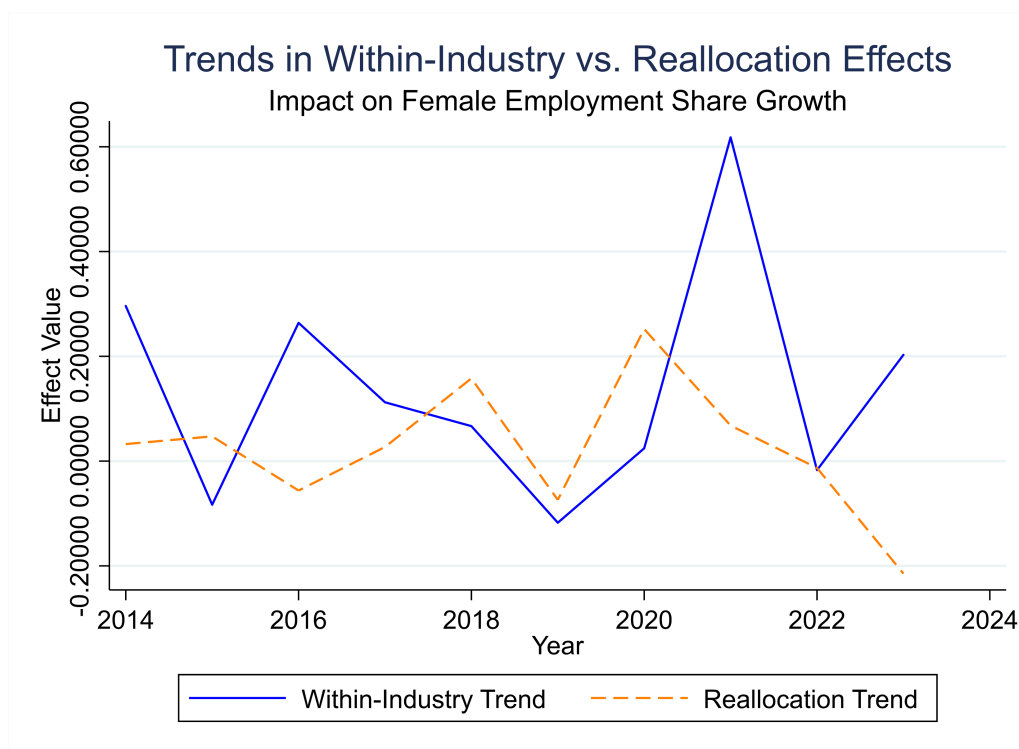


Figure 1: Decomposition of female employment-share growth in organised manufacturing into within-industry and reallocation effects, 2014–2023.

The within-industry effect is volatile across the period and remains positive in most years. In 2014 a strong positive value indicates that female employment expanded more rapidly than total employment within existing industries. The effect turned negative in 2015, when female employment growth lagged overall employment growth. A recovery followed over 2016–2018, though the magnitude weakened year by year. In 2019 the within-industry component again became negative, implying a deterioration in women’s relative employment opportunities inside industries. A sharp increase in 2021 suggests that the rise in the female employment share that

year owed mainly to changes within industries rather than to structural shifts across them. That increase may reflect differential employment adjustment after the COVID-19 pandemic, with male employment contracting more rapidly or female employment recovering relatively faster. The positive within-industry effect cannot, however, be read as an unequivocal improvement in women’s labour-market outcomes, since a rising share may also follow from disproportionate job losses among men.

The reallocation effect stays small throughout, indicating that changes in the industrial composition of employment contributed only modestly to changes in the female employment share. Positive reallocation effects in 2017–2018, and especially in 2020, suggest that employment growth was concentrated in industries with above-average female intensity. The negative values in 2016, 2019, and 2023 indicate the opposite: employment expansion concentrated in relatively male-intensive industries, or contraction concentrated in industries with higher female participation.

Comparing the two components, the within-industry effect is consistently larger in magnitude than the reallocation effect. Changes in women’s employment shares in organised manufacturing have thus been driven primarily by processes occurring inside industries—technological change, labour-saving innovation, shifting skill requirements, and gender-specific hiring—rather than by structural transformation moving employment across industries. This is consistent with Tejani and Kucera (2021), who find within-industry effects to be the dominant source of change in female manufacturing employment shares. The limited contribution of the reallocation effect further implies that structural transformation has not been oriented toward industries with higher female employment intensity. The trajectory of women’s manufacturing employment therefore turns on industry-specific factors and technological upgrading more than on the reallocation of labour between industries.

5.2 Capital intensity and the female employment share

Figure 2 plots the relationship between capital intensity—measured as the logarithm of fixed assets per worker—and the female share of total employment across manufacturing industries. Each point is an industry–year observation; the fitted line summarises the overall association.

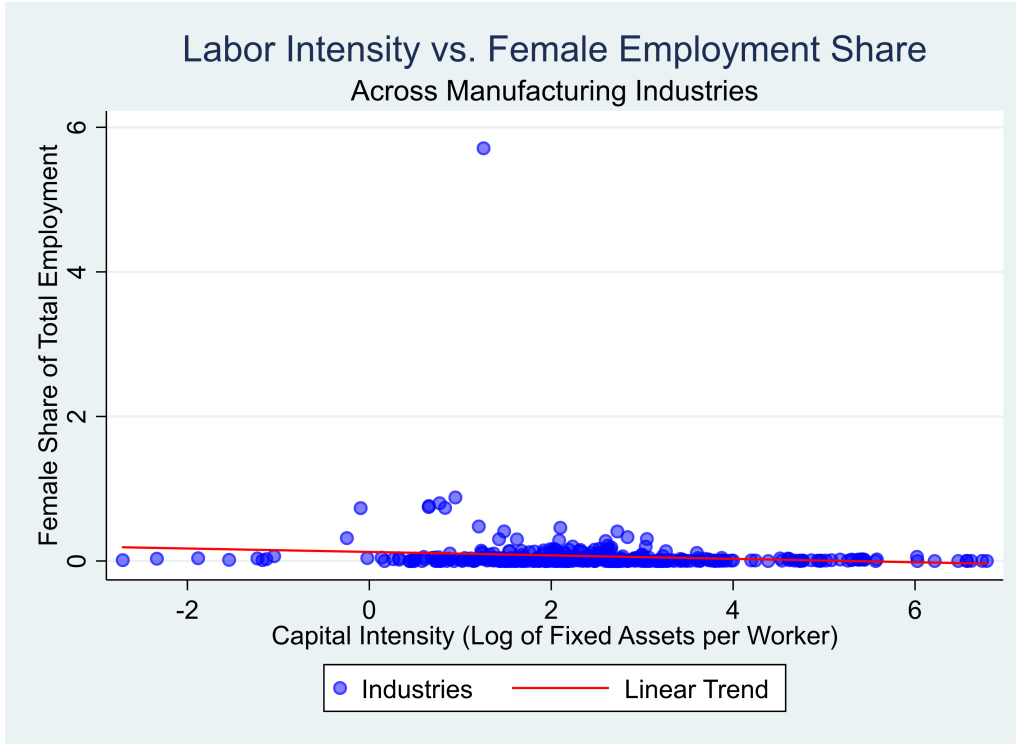


Figure 2: Capital intensity (log of fixed assets per worker) and the female share of total employment across manufacturing industries, with fitted linear trend.

The scatter implies a weak negative relationship between capital intensity and the female share of employment. As industries become more capital-intensive, the proportion of women in total employment tends to fall, though the association is not strong. The downward slope of the fitted line indicates that industries with higher capital per worker are generally those with lower female employment shares.

A striking feature is the clustering of observations at low female shares across a wide range of capital-intensity levels: in most manufacturing industries women make up a small proportion of total employment regardless of how capital-intensive the industry is. The decline in the fitted line suggests that technological upgrading and capital deepening have not translated into greater employment opportunities for women. The wide dispersion, particularly among industries of moderate capital intensity, points to considerable heterogeneity: some industries sustain relatively high female shares at moderate capital intensity, while others record very low female participation at comparable levels. Factors such as skill requirements, occupational segregation, the technological character of production, labour-market institutions, and gender-specific hiring practices evidently also shape women's participation. A handful of outliers display unusually high female shares, indicating that certain industries remain female-intensive and may have production structures more accommodating to women's employment; these cases are exceptional and do not alter the overall pattern.

Capital deepening in manufacturing, in short, is not accompanied by a corresponding rise in women's employment. Higher capital intensity is associated, if anything, with lower female employment shares. This accords with the literature on structural transformation and gendered employment, which holds that technological change and rising capital intensity tend to favour

skill-intensive processes and occupations traditionally dominated by men, thereby limiting the absorption of female labour. Industrial modernisation alone is therefore insufficient to deliver gender-inclusive employment growth.

6 Discussion

Two findings stand out, and they reinforce one another. The decomposition shows that the female employment share moves mainly because of what happens *inside* industries, not because employment reallocates toward or away from female-intensive ones. The cross-section shows that the industries absorbing more capital tend to employ proportionally fewer women. Both point to the same conclusion: the forces reshaping women’s place in organised manufacturing operate at the level of the production process—who is hired and retained as technology, skill requirements, and work organisation change—rather than at the level of which industries grow.

The muted reallocation effect carries its own message. Had structural transformation been pulling employment toward female-intensive industries, the reallocation component would have been consistently positive and sizeable. Instead it is small and changes sign, turning negative in 2016, 2019, and 2023. India’s organised manufacturing has not been expanding disproportionately in the activities where women are most present; in several years it has done the reverse. The cross-sectional pattern in Figure 2 is the static counterpart of this dynamic result. A negative slope of capital intensity against the female share is what one expects if the same technology-linked forces that drive the within-industry effects also sort women out of the more capital-intensive industries. The reading is consistent with Tejani and Milberg (2016), who find capital intensity rather than export growth to be the operative margin, and with the upgrading-and-defeminization mechanism documented by Tejani and Kucera (2021) and Greenstein and Anderson (2017).

The 2021 spike in the within-industry effect deserves a cautious interpretation. It is large enough to dominate the series, but the methodology itself warns against reading a rising female share as straightforward progress: the same positive value would appear if male employment fell faster than female employment during the pandemic’s disruption. The decomposition tells us *where* the change in the female share originated—inside industries—but not whether it reflects women gaining or men losing. On the evidence here, the post-2020 movement is better described as a compositional effect of an unusual labour-market shock than as evidence of durably improving opportunities for women.

Read as a whole, the results temper the expectation that industrial growth will mechanically broaden women’s economic inclusion. Manufacturing has expanded and modernised, but the gender composition of its workforce is governed by industry-internal dynamics that capital deepening does not appear to soften, and may sharpen.

7 Conclusion

This paper has asked whether changes in women’s presence in India’s organised manufacturing are driven by shifts within industries or by the reallocation of labour across them, and has answered the question using an annual decomposition of the female employment share for 2014–2023, complemented by the cross-sectional relationship between capital intensity and the female share. The within-industry effect dominates and is volatile; the reallocation effect is small and occasionally adverse; and more capital-intensive industries tend to employ proportionally fewer women. Together these findings indicate that structural transformation in organised manufacturing has not been oriented toward female-intensive industries, and that capital deepening has not, by itself, widened women’s access to manufacturing employment.

The policy implication follows directly. Because the binding margin lies inside industries and is bound up with technology and skill, expanding manufacturing output or upgrading its capital stock will not on its own make employment more gender-inclusive. What the evidence points to instead is targeted action on the within-industry channel: building women’s skills in the occupations that capital-intensive production demands, dismantling the occupational segregation that confines women to a narrow band of low-value-added industries, and improving their access to technology-intensive work. Industrial and social policy can be designed so that labour demand becomes more gender-equalising rather than less (Seguino and Braunstein, 2018). Without such measures, the gains from industrial modernisation are unlikely to reach women and men on equal terms.

8 Limitations and Future Work

Several limitations qualify these conclusions and mark out where further work is needed. First, the analysis covers organised (registered) manufacturing only. The ASI frame excludes the unorganised sector, where a large share of India’s women workers are employed, so the results speak to formal manufacturing rather than to manufacturing as a whole. Second, the decomposition is built on employment *shares*, and—as the methodology itself flags—both the within-industry and reallocation effects can move for reasons unrelated to genuine improvement for women, most notably faster male job loss. The decomposition locates where change occurs; it does not by itself identify whether that change represents a welfare gain. Third, the exercise is descriptive: it distinguishes within- from between-industry change and reports a cross-sectional association with capital intensity, but it does not isolate the causal contributions of technology, skill requirements, and social norms, which the literature treats as intertwined. Fourth, the study window includes the COVID-19 shock, which complicates the interpretation of the 2020–2021 values in particular.

Future work could extend the analysis in several directions: linking the employment decomposition to data on wages, skill, and contract type to test whether reallocation toward female-intensive industries coincides with better or worse job quality; disaggregating the decomposition by industry group to identify which activities drive the aggregate pattern; extending coverage to the unorganised sector through the Annual Survey of Unincorporated Sector Enterprises or

the Periodic Labour Force Survey; and pursuing causal identification of how capital deepening affects female labour demand. Each would sharpen the picture this paper has drawn of where, and how, women are being reallocated within Indian manufacturing.

References

- Barrientos, S., Kabeer, N., and Hossain, N. (2004). The gender dimensions of the globalization of production. Working paper, International Labour Office, Geneva.
- Caraway, T. L. (2007). *Assembling Women: The Feminization of Global Manufacturing*. Cornell University Press, Ithaca, NY.
- Elson, D. and Pearson, R. (1981). ‘nimble fingers make cheap workers’: An analysis of women’s employment in third world export manufacturing. *Feminist Review*, 7(1):87–107.
- Greenstein, J. and Anderson, B. (2017). Premature deindustrialization and the defeminization of labor. *Journal of Economic Issues*, 51(2):446–457.
- Joekes, S. (1999). A gender-analytical perspective on trade and sustainable development. In *Trade, Sustainable Development and Gender*. United Nations Conference on Trade and Development (UNCTAD), Geneva.
- Kucera, D. and Tejani, S. (2014). Feminization, defeminization, and structural change in manufacturing. *World Development*, 64:569–582.
- Mitra, S. (2026). Women’s employment and occupational segregation in indian manufacturing. Forthcoming; full bibliographic details to be confirmed by the author.
- Ocampo, J. A., Rada, C., and Taylor, L. (2009). *Growth and Policy in Developing Countries: A Structuralist Approach*. Columbia University Press, New York.
- Seguino, S. (1997). Gender wage inequality and export-led growth in south korea. *Journal of Development Studies*, 34(2):102–132.
- Seguino, S. (2000). Gender inequality and economic growth: A cross-country analysis. *World Development*, 28(7):1211–1230.
- Seguino, S. and Braunstein, E. (2018). The role of manufacturing and industrial policy in promoting gender equality. In *Rethinking Economic Policies for Inclusive Development*. United Nations Conference on Trade and Development (UNCTAD), Geneva. Bibliographic details to be confirmed by the author.
- Standing, G. (1989). Global feminization through flexible labor. *World Development*, 17(7):1077–1095.
- Standing, G. (1999). Global feminization through flexible labor: A theme revisited. *World Development*, 27(3):583–602.

- Tejani, S. and Kucera, D. (2021). Defeminization, structural transformation and technological upgrading in manufacturing. *Development and Change*, 52(3):533–573.
- Tejani, S. and Milberg, W. (2016). Global defeminization? industrial upgrading and manufacturing employment in developing countries. *Feminist Economics*, 22(2):24–54.